

Kubernetes ReplicaSet

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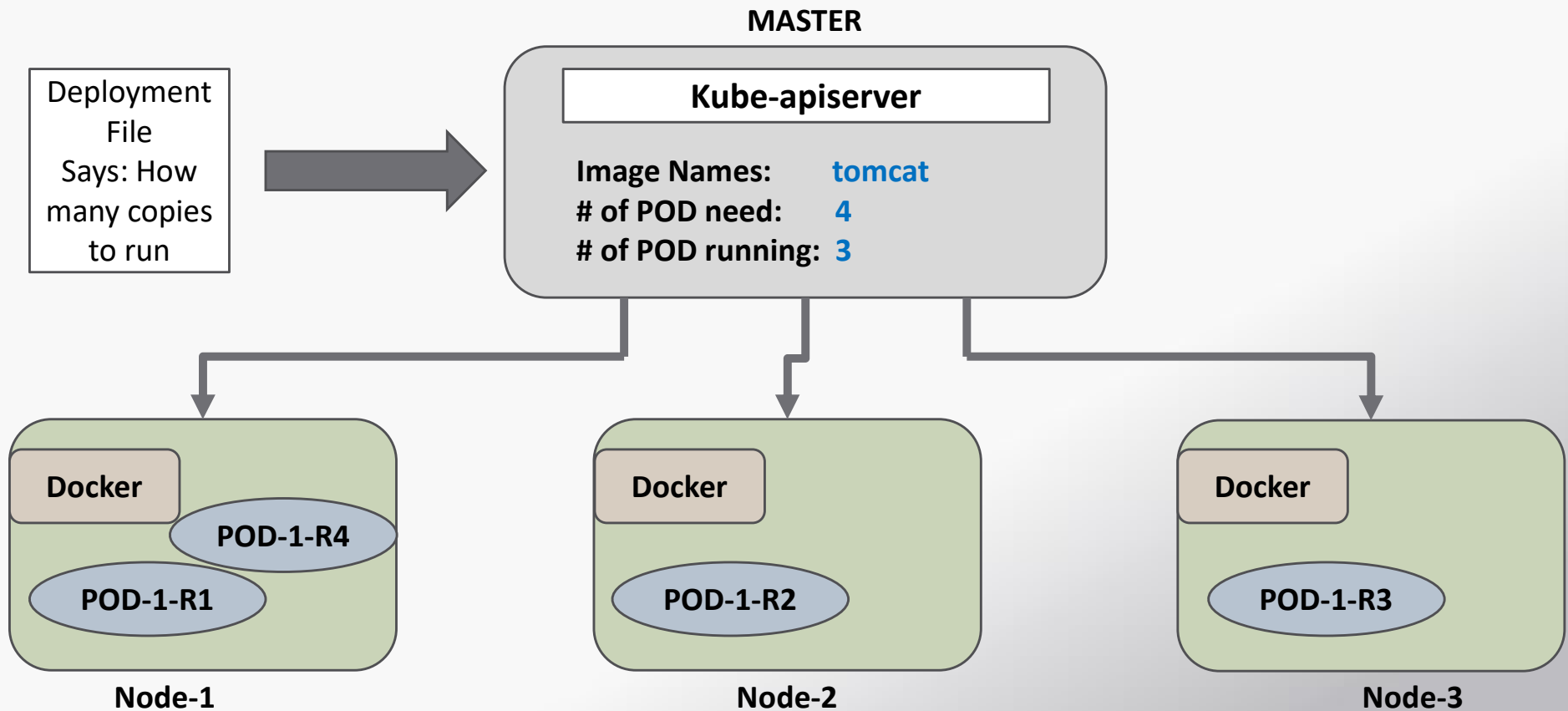


Agenda

- Kubernetes Working
- Overview – ReplicaSet
- Labels & Selectors
- Equality-based and Set-based Selectors



Kubernetes Working





ReplicaSet

- Ensures that a specified number of pods are running at any time
- If there are excess pods, they get killed to come back to same number of pods.
- Or if there is less (due to deletion or termination or host loss), then the pods will get created to match the number.
- This is taken care with the help of “LABELS”



ReplicaSet vs Replication Controller

ReplicaSet	Replication Controller
Its next-Gen	Previous Generation
This uses “Set-Based Selectors”	This uses “ Equality-based Selectors
Operators used → in notin exists	Operators used → = == !=
... selector: matchlabels: app: nginx tier: backend	... selector: app: nginx tier: backend ...



Labels & Selectors

- Labels are key-value pair

apiVersion: v1

Kind: pod

Metadata:

name: nginx-pod

labels:

app: logic

tier: backend

spec:

replicas: 3



Using selectors we can refer to this label to map to this pod



Equality-based Selectors with example

Operators: $\rightarrow$ = == !=

Examples:

- environment = production
- tier != backend

Command line

- `$ kubectl get pods -l environment=production`

In manifest

-
- **selector:**
- environment: production
- tier: backend
-

Supports:

Services,
Replication Controller



Set-based Selectors with example

Operators: → in notin exists

Examples:

- environment in (production, staging)
- tier notin (backend)

Command line

- \$ kubectl get pods -l environment in (production)

In manifest

-
- selector:
- matchExpressions:
- - {key: environment, operator: in, values: [production, staging]}
- - {key: tier, operator: notin, values: [backend]}
-

Supports:

Replica Set

Job

Deployment

Daemon Set



De-coding the YAML file

```
apiVersion: v1
```

```
kind: ReplicaSet
```

```
metadata:
```

```
  name: tomcat-rs
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: tomcat-app
```

```
    matchExpressions:
```

```
      - {key: tier, operator: In,  
values: [frontend]}
```

```
template:
```

```
  metadata:
```

```
    name: tomcat-pod
```

```
    labels:
```

```
      app: tomcat-app
```

```
      tier: frontend
```

```
  spec:
```

```
    containers:
```

```
      - name: tomcat-cont
```

```
        image: tomcat
```

```
        ports:
```

```
          - containerPort: 8080
```



De-coding the YAML file

apiVersion: v1

kind: ReplicaSet



Define the kind of Object as Replicaset

metadata:

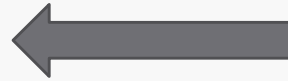
name: tomcat-rs



Name for the ReplicaSet

spec:

replicas: 3



Number of replicas of the POD needed

selector:

matchLabels:

app: tomcat-app

matchExpressions:

- {key: tier, operator: In,
values: [frontend]}



De-coding the YAML file

```
apiVersion: v1
kind: ReplicaSet
metadata:
  name: tomcat-rs
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: tomcat-app
```

```
    matchExpressions:
```

```
      - {key: tier, operator: In, values: [frontend]}
```



This is to select the container that needs to be replicated.

MatchLabel : match's the container custom label

MatchExpression:

It's a list of pod selector requirements.



De-coding the YAML file

```
template:
  metadata:
    name: tomcat-pod
    labels:
      app: tomcat-app
      tier: frontend
  spec:
    containers:
      - name: tomcat-cont
        image: tomcat
        ports:
          - containerPort: 8080
```



This is to select the container that needs to be replicated.

MatchLabel : match's the container custom label

MatchExpression:
we